velociraptor: exact light curves and GPU-accelerated false-positive probabilities for TESS planet candidates

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Abstract

TRICERATOPS computes the probability that a transit candidate is a false positive by averaging likelihoods over prior draws. We present velociraptor, a GPU implementation of its three likelihood functions that leaves the draws, the priors and the probabilities unchanged. The transit light curve is the closed form of Agol, Luger and Foreman-Mackey on the exact Kepler orbit, in Triton kernels with a single-precision first pass. It agrees with 50-digit references to 2.78×10^{-16} in the lens area and 6.34×10^{-14} in the orbit with NumPy 1.26.4. The light curve of PyTransit 2.2.0, which TRICERATOPS uses, departs from a 50-digit reference by 4.5×10^{-4} to 4.1×10^{-3} on the draws that carry the evidence of some short, eccentric orbits, outside the range in which its authors tested it. On 158 search candidates, the exact model shifts the false-positive probability (FPP) or the nearby-star probability by at least one seed scatter on 15, and moves FPP across a decision value on 2. At 10^6 draws the shifts persist on three of four stars. One calculation of 10^5 draws takes 0.4 to 0.8 s on a consumer card, against 1.5 to 5.5 s for the published code on 32 CPU threads; 20 runs of 10^6 draws take 88 to 105 s, against an extrapolated 661 to 891 s. The package also reports the scatter of the probabilities between seeded batches. Out of sample it covered the truth in 0.667 of trials on six stars, so the package states no coverage for it.

1 Introduction

A transit candidate is validated statistically when the probability that it is a false positive is small. TRICERATOPS [1] computes this probability for a target from the evidence of a set of scenarios: the target hosts a transiting companion, the target or a neighbour hosts an eclipsing binary, a background star does, and so on. The evidence of each scenario is a mean of the likelihood of the light curve over draws from the prior of the scenario's parameters. The probabilities FPP and NFPP follow from the normalised evidences. A candidate is validated when FPP is below 0.015 and NFPP below 0.001 [1]. TRICERATOPS validated 12 objects of interest in a vetting of 384 TESS objects of interest [1] and 64 further TESS planets in a recent survey of 443 candidates [10]. Other tools compute such probabilities by other means: VESPA [2, 3] simulates a population of instances of each false-positive scenario, PASTIS [4] compares hypotheses by a Bayesian model comparison, and BLENDER [5] models the photometry as blend scenarios.

The light curve of the transit must be right on the orbits of the draws, including short and eccentric ones. The speed must fit the published practice: Collier et al. [10] run the calculation 20 times per target, each run with its own seed and 10^6 draws, and Giacalone et al. [1] chose $N = 10^6$

as the number of draws that gives results consistent between consecutive calculations (their Section 2.4) and repeated the calculation 20 times for each of their 12 validated objects (their Section 6). The Monte Carlo also needs a statement of how far it can be trusted, since the scatter of FPP between seeds depends on the star (Section 6).

velociraptor replaces the three likelihood functions of TRICERATOPS 1.1.0 with a calculation on a graphics processor and leaves the draws, the priors and the probabilities unchanged. Its transit light curve is the closed form of Agol, Luger and Foreman-Mackey [7] on the exact Kepler orbit. Section 2 describes the design, Section 3 the accuracy against references, Section 4 the agreement with the published code and the stars on which they differ, Section 5 the time and memory, and Section 6 the scatter between seeded batches that the package reports and what was measured about it. Section 8 lists what was not verified.

2 Design

2.1 The calculation

For each scenario TRICERATOPS draws N parameter vectors from the prior, computes a model light curve for each, forms the chi-squared against the folded light curve, and takes the evidence as the mean of $\exp(-\chi^2/2)$ times the prior-volume factors of the scenario. The light curve is a quadratic limb-darkened transit [6] on a Keplerian orbit, integrated over the exposure. Throughout, the published code is TRICERATOPS 1.1.0 with its default settings and its transit model, PyTransit 2.2.0 [8].

2.2 The exact model

The flux removed by the planet is the integral of the stellar intensity over the part of the planet's disc inside the star. velociraptor evaluates it by the closed form of Agol, Luger and Foreman-Mackey [7], which writes the integral with complete elliptic integrals, and by a second implementation after Mandel and Agol [6], a Gauss–Legendre quadrature of the arc of the planet's disc inside each ring of the star, used as a cross-check. Where the planet is small and wholly inside the stellar disc, which is nearly every element of a calculation, the flux is a series in the radius ratio, the mean of the intensity over the planet's disc (Pizzetti's series for the mean value of an analytic function), of order 3 or 9 chosen from a bound on the first omitted term, so that no elliptic integral is evaluated there. The orbit is the exact Kepler orbit: Kepler's equation is solved for each point of each exposure, with no Taylor expansion about mid-transit.

2.3 The device kernels

The kernel is written in Triton and runs on ROCm and CUDA devices. Each lane handles one pair of a draw and a point: Kepler's equation and the sky separation for each exposure sub-sample, then the flux. The complete elliptic integrals are evaluated by Bulirsch's algorithm in at most 8 steps of the Landen recurrence, left as soon as every lane of the block has converged. A block of 32 lanes with no sub-sample over the stellar disc skips the closed form.

The kernel has two passes. The first pass forms, for each exposure node, the small change of the eccentric anomaly from the centre of the exposure (a few Newton steps on an angle of at most

0.1 rad) and the order-3 series flux of a small planet wholly inside the disc, as a deficit, in single precision. It flags a pair for the second pass when a node is not on the series (ingress and egress, a larger planet, the band within 10^{-4} of the contact $b + k = 1$) or when the exposure moves the anomaly by more than 0.1 rad. The orbit of the centre, the weights, the tests, the sum into the flux and the second pass, the closed form for the flagged pairs, are in double precision. A guard at the outer contact keeps a lane that lies just inside the contact in double precision from rounding outside it in single precision (Section 3.3).

The kernel is launched in slices of at most 2^{20} sub-sample elements, so that no dispatch runs long enough to starve the display or trip the driver's ring timeout; the slice is a multiple of 32 pairs, so the blocks, and with them the last bits of the flux, do not depend on the slice size.

The exposure is integrated with 3 Gauss–Legendre nodes by default and with the 20 equal parts of the published code as an option. The default is a different quadrature from the published code's, with its own difference from the 20 midpoints (Section 3.3). Throughout this paper the default is the call `install("cuda")` with no further arguments: the Triton kernel, 3 exposure nodes and the single-precision first pass, with the overlap of 3 scenario calls.

2.4 Integration with TRICERATOPS

`install` puts the package's functions in place of TRICERATOPS's three likelihood functions for the length of a block and puts them back on leaving it. Three further pieces handle the host.

First, the splines of TRICERATOPS that sample a scenario hold the interpreter lock through scipy's evaluation, which prevents the sampling of one scenario from overlapping that of the next. velociraptor evaluates them with the recurrence of de Boor, with the same operations in the same order, compiled without the lock; the values are the same to the last bit. Second, the overlap: `calc_probs` calls its scenario functions in a fixed order and each draws all its random numbers before its first likelihood call, so the sampling of the next scenario can begin once the likelihood of the one before has begun. Inside the overlap block `calc_probs` runs once with stand-ins that record the calls, and then for real with the recorded calls run by worker threads in order; the draws, their order and the results are those of the sequential run bit for bit, and the overlap is given up and the sequential run done if the recorded and real calls differ. Third, TRICERATOPS draws every scenario from one global random stream, so leaving one scenario out changes the numbers of all the others; `scenario_streams` gives each call a stream fixed by the seed, the function and the index of the call.

The seeded batches of Section 6 seed numpy's one global generator for each batch, which concurrent calls would share. The package therefore runs the calls of a batch one at a time and suspends the overlap inside `batches`, so batches cost the time that the overlap saves.

2.5 Stars

Table 1 lists the stars used and the rule that chose each set. The criteria of each comparison of Sections 4 and 6, what counts as a shift, a change of side, a persistence or a coverage, and the hypotheses tested, were fixed before its measurements were made. The stars of the screen set are TESS targets on which a periodic transit search of the light curve found a signal above its detection threshold and which were then passed to TRICERATOPS. They are candidates of a search and

not a random sample of stars.

Table 1: The stars used. A hash order is the ascending order of the SHA-256 of a fixed string and the TIC identifier of the star.

Set	stars	Selection rule	Used in
Precision and speed stars	4	TIC 396212145, 206361339, 42071226 and 201354238, the stars of the benchmark of the package, with no selection rule beyond the availability of their stored tables; the first two carry the tolerance test.	Sections 3, 5
Screen set	159	The stars with a stored TRICERATOPS table in earlier calculations: 45 flagged by a screen (the best draw of a scenario of probability at least 0.1 has eccentricity above 0.5 and scaled semi-major axis below 15), 113 not flagged, 1 unscreened. Each stratum is taken in a hash order. The comparison of the two models of Section 4.1 ran on 158 of them, the 45 flagged and all 113 not flagged; one further star was excluded because TRICERATOPS refused it, and one is unscreened and was not run.	Section 4.1
Coverage stars	6	The first six stars in hash order whose FPP scatter over ten seeds exceeds 10^{-3}, excluding the four stars of the in-sample coverage measurement; 100 batches of 10^5 draws.	Section 6

3 Accuracy

3.1 Against references

Against references that do not come from the package the closed form differs by at most 2.78×10^{-16} from the lens area in 50 digits without limb darkening (552 cases), 1.52×10^{-13} from a two-dimensional integral (10 cases) and 6.44×10^{-15} from 768 nodes of the quadrature (3,312 cases), and the sky separation of an orbit differs by at most 6.34×10^{-14} with NumPy 1.26.4 from a 50-digit solution (9 cases).

3.2 The exposure integral

In these four calculations the 3-node default agrees with the integral of 64 nodes at the same seed: its FPP differs by at most 9.1×10^{-6} and its NFPP by at most 1.0×10^{-5}, which is at most 0.003 of the scatter of FPP or NFPP between ten seeds of the 3-node default on each star.

3.3 Single precision

The first pass. The mixed-precision kernel forms the times relative to mid-transit, the flux as a deficit, sums the chi-squared in double precision and evaluates the closed form near contact in double precision. A second variant evaluates the closed form in single precision as well, with a guard at the outer contact. The flux of the closed form in single precision differs from double precision by at most 3.6×10^{-7} over radius ratios 0.003 to 1.7 and four limb-darkening laws, and by at most

3.3×10^{-7} on the nodes of a planet row, against the rule of 10^{-6}. Without the guard, a lane just inside the outer contact in double precision can round outside it in single precision, and the flux is wrong by up to 9. The cause is rounding across a branch and not a loss of precision in the formula.

Shift of the probabilities. Table 2 gives the largest shift in any $\ln Z$ and the shift in FPP of the single-precision first pass against double precision, at 20 midpoints, on four stars. The shifts of FPP are 5.6×10^{-7} to 4.6×10^{-6} where FPP is not pinned. On the two stars with a stored seed scatter the smallest scatter of the ln probability of a scenario is 7.2 and 0.7, against largest shifts of 0.0185 and 2.5×10^{-4}. No probability changes side of 0.015 or 0.5.

Table 2: Shift of the evidence and of the probabilities from a single-precision first pass against double precision, whole calculation at the 20-midpoint exposure integral.

TIC	largest shift of $\ln Z$	shift of FPP	shift of $\ln$ NFPP
396212145	0.0185	0 (FPP 1)	5.4×10^{-5}
206361339	2.5×10^{-4}	4.6×10^{-6}	6.4×10^{-5}
42071226	8.8×10^{-6}	5.6×10^{-7}	1.5×10^{-6}
201354238	2.5×10^{-5}	1.9×10^{-6}	(NFPP 0)

The default of the package is the 3 nodes with the single-precision first pass, and it was checked as a default on the four stars of Table 4, each at eight seeds of 100,000 draws, against the same calculation in double precision at 3 nodes. The largest shift of any $\ln Z$ is 0.32 percent of the seed scatter of its row, against a limit of 10 percent; the sides of FPP = 0.015 and 0.5 and of NFPP $= 10^{-3}$ agree at every seed; the largest differences are 1.3×10^{-4} in FPP and 1.1×10^{-4} in NFPP, both on one star. On one star three of 18 rows are $-\infty$ in both precisions at every seed and carry no test. The default is 1.09 to 1.59 times faster than double precision at 3 nodes (Table 4).

A kernel in single precision only. The single-precision closed form is measured as a function of the radius ratio, the limb darkening and the distance from contact, with the largest error given above. Whether a kernel with no double precision at all meets the rule of 10^{-6} in flux is not settled. On the benchmark's draws the error is within it up to periods of about 3 days and reaches 3.5×10^{-6} at 30 days, growing with the semi-major axis in stellar radii, which points at the rounding of the orbit. Such a kernel is not built.

3.4 The precision the probabilities need

The probabilities are insensitive to errors in the flux of the size that single precision makes. On two stars, one with FPP pinned at 1 and one with FPP 0.090, we added an error to the light curve and found the largest error at which every scenario whose scatter between ten seeds exceeds 10^{-3} shifts by less than a tenth of its scatter, and $\ln$ NFPP likewise. An absolute flux error that is independent for each element is tolerated at 10^{-7} and not at 10^{-6}. A relative error in the deficit $1 - F$ is tolerated at 10^{-3} and not at 10^{-2}. At the largest tolerated level the largest shift is 0.11 of the limit for the flux (at 10^{-7}) and 0.25 for the deficit (at 10^{-3}); at the next level it is 1.07 and 2.46. The FPP of the star at 0.090 moves by 6.3×10^{-3} under a flux error of 10^{-6} and by 9.5×10^{-4} under a deficit error of 10^{-3}, against its scatter of 0.17. Rounding the flux to single precision shifts

the probabilities by 0.004 of the limit, and rounding the deficit by 3.5×10^{-6} of it. The rounding is a deterministic function of the flux and not an independent error, and its shift is smaller than that of an independent error of the same size.

The tolerance holds for independent errors. A systematic error, such as the bias of an orbit approximation that is the same on every draw, adds over the points of the light curve and has a lower tolerance. It is also relative to the scatter at 10^5 draws, and a run with more draws has a smaller scatter and a tighter tolerance.

4 Agreement with the published code, and where they differ

The two calculations draw the same parameters and form the same chi-squared, so the probabilities differ only where the two light curves differ on the draws that carry the evidence of a call. On the heaviest draw of the control call of a star that did not shift the two curves agree, and so do the probabilities. Where the heaviest draw of a call is a draw on which the curves differ, FPP may move; agreement on the published code's heaviest draw is not enough, because the exact calculation may weight another draw most, as on TIC 42071226 (Section 4.1).

4.1 Where the published transit model differs

The light curve of PyTransit 2.2.0, which TRICERATOPS 1.1.0 uses, departs from the exact light curve on some orbits. Section 2 gives the exact orbit and the closed form of the package. We swept a grid of 320 parameter sets, not drawn from a population: four periods (0.5, 0.713, 2 and 10 d), five scaled semi-major axes (3, 4, 6, 10 and 20), four eccentricities (0, 0.3, 0.5 and 0.7) and four arguments of periastron. In 70 of the 320, PyTransit's model puts a second dip, a flux under $1 - 10^{-6}$ where the exact flux is 1, at 0.004 to 0.25 of the period from mid-transit: in none of the 80 sets at zero eccentricity, 10 of the 80 at 0.3, 24 at 0.5 and 36 at 0.7. The flux of the model is off the exact one by more than 10^{-5} in 143 of the 320.

The validation of the package compares the two models on 300 further orbits, drawn at random with periods of 0.5 to 30 d, $a/R_\star$ of 2 to 40 and eccentricities of 0 to 0.9. For $a/R_\star$ of 20 to 40 and eccentricities under 0.3 the median of the largest flux difference of an orbit is 5.0×10^{-9}. At eccentricities over 0.6 the largest difference is 0.078 for $a/R_\star$ of 5 to 10 and 0.17 for the one orbit with $a/R_\star$ under 3.

One orbit of that comparison gives a minimal reproduction. Take $P = 21.99247$ d, $a/R_\star = 6.964572$, an inclination of $87.04449°$, $e = 0.8502341$, $\omega = 347.7677°$, $k = 0.4962705$, quadratic limb darkening $(u_1, u_2) = (0.4, 0.25)$ and mid-transit at $t = 0$. At $t = -0.7894$ d, with a 30-minute exposure evaluated at its centre, the `QuadraticModel` of PyTransit 2.2.0 with `interpolate=False` gives a flux of 0.92212, where the planet is off the stellar disc and the exact flux is 1.

The size of the shift in FPP. On four stars, the FPP of the published code differs from that of the exact calculation on the same draws by 5.8 and 7.7 seed scatters on two of them (1.0000 against 0.9024, and 0.7376 against 0.7725) and agrees on the other two. Four stars give no rate, so we compared the two models on the 158 stars of the screen set of Table 1 that were run: 45 flagged by the screen and 113 not flagged, in hash order within each stratum. Each calculation, with 100,000 draws, ran under both models on the same draws at three seeds, and under the exact

model at ten seeds, whose standard deviation of FPP is the seed scatter. The exact model ran in double precision with the 20 exposure midpoints of the published code, so the two calculations differ in the light curve alone and not in the quadrature or the precision. Appendix A lists the stars with their FPP under both models. A star is shifted in FPP if the mean difference of FPP between the two models is at least the larger of that scatter and 0.01 and has that size and sign at each of the three seeds; in $\ln$ NFPP the same with a floor of 0.095, counted only where NFPP is at least 10^{-3} in one of the models.

Of the 45 flagged stars 5 are shifted in FPP (11 percent, Wilson 95% interval 4.8 to 23.5 percent) and none in $\ln$ NFPP. Of the 113 unflagged stars, which are all the unflagged stars of the set, 10 are shifted (8.8 percent, 4.9 to 15.5 percent), 6 in FPP and 5 in $\ln$ NFPP, one in both. Over the 158 stars 15 are shifted (9.5 percent, 5.8 to 15.1 percent). The interval of the unflagged share includes the point value of the flagged share, so the screen does not separate the stars that shift. Over the 11 stars shifted in FPP, the published code's FPP is higher on 8 and lower on 3 (a two-sided sign test gives p of about 0.23); the direction is observed here, and no direction is claimed in general. The hypothesis that stars of low noise per point shift more is not supported: 3 of 50 stars shifted in the lowest third of the noise and 8 of 50 in the highest (one-sided Fisher $p = 0.97$, on the first 150 stars run).

Two stars of the 158 change side of a decision value between the two models (1.3 percent, Wilson 0.35 to 4.5 percent), with the FPP of the published code the lower in both, and none changes side of the NFPP line of 10^{-3}. Fewer than three stars that change side are stated as cases and not as a rate of changes of class; these are two cases.

The first, TIC 325177252, changes side of FPP = 0.015. It is not flagged and has a noise per point of 5.0×10^{-5}. The FPP of the published code is 0.00028, 0.00020 and 0.0021 at three seeds, all under 0.015, and that of the exact model is 0.026 to 0.038 at ten seeds (scatter 0.0037), all over it; NFPP is under 10^{-3} in both. On the criteria of TRICERATOPS itself, the published code would pass the star and the exact model would not.

The second, TIC 424717049, changes side of 0.5 and is the largest shift, 83 seed scatters. It is not flagged and has a noise per point of 2.4×10^{-3}. The FPP of the published code is 0.313, 0.0002 and 0.0001 at three seeds and that of the exact model 0.858 to 0.888 at ten. At seed 1 the published code puts 0.62 of the probability on the scenario PTP, 0.31 on STP and 0.06 on the target's planet (TP), while the exact model puts 0.54 on SEB and 0.08 on TP; the scenario names are those of TRICERATOPS.

One further star, TIC 278659026, differs by 23 seed scatters (the published code 1.0, exact model 0.577, both above 0.5).

Against a reference in 50 digits. On each of the eight shifted stars replayed we took the call with the largest difference of $\ln Z$ between the models and computed the light curve of the draw that each model weights most against a reference built from Kepler's equation in 50 digits. On the four flagged stars of the first part, the exact curve of velociraptor differs from the reference by at most 2.2×10^{-16} on all eight draws. The curve of the published code differs by 4.5×10^{-4} to 3.0×10^{-3}, against a noise per point of 3.5×10^{-4} to 1.8×10^{-3}, and half its chi-squared against the data is lower than the reference's by 11 to 32 on the draw that the published code weights most. On two of the four stars the curve of the published code is also off by 8.0×10^{-4} and 4.5×10^{-4}

on the draw that the exact calculation weights most, and the data do not prefer it there. On TIC 325177252 the exact curve is within 3.3×10^{-16} of the reference, and the published code is off by 4.1×10^{-3}, 80 times the noise, on the draw that both models weight most. On TIC 278659026 the heaviest draws lie in the screened region ($k = 0.76$, $e = 0.70$, $a/R_\star = 6.7$). There the published code is off by about 0.0029 in the flux of the target, against a fold error of 0.0019. The host star of the scenario is a faint neighbour, whose flux and error the package rescales by about 144, and in its flux the error is 0.42 against a fold error of 0.278. The exact curve is within 2×10^{-15} of the reference, and the heaviest draw of the published code lies at the 99.99th percentile of Δ. On TIC 42071226 the heaviest draw of the published code agrees with the reference to 3×10^{-9}. On TIC 424717049, call 6 (the STP row, $\ln Z$ differing by +47.4), the heaviest draw of the published code ($k = 0.095$, $e = 0.39$, $a/R_\star = 2.26$, inclination 86.0°) is off the reference by 2.15×10^{-3}, 0.9 of the noise per point, and the exact curve is within 2.2×10^{-16}; half the chi-squared against the data is 424.7 for the published code and 480.8 for the reference. The secondary eclipse of the eclipsing-binary likelihoods is not part of the reference.

What makes the shift. The evidence of a call rests on its heaviest draws. For every draw of the replayed calls on which the two curves differ by more than 10^{-6} we computed the difference Δ of half the chi-squared of the exact curve and of the published code's curve (positive when the published code fits better). The exact curve stands in for the 50-digit reference, from which it differs by at most 4.4×10^{-15} on the heaviest draw and ten affected draws of each call. Each pattern is read from the heaviest draw of the call with the largest difference of $\ln Z$; the direction of the shift in FPP follows from the pattern and from the scenario to which the call belongs.

The first pattern is a favourable error on the heaviest draw. The heaviest draw of the published code is one on which its error happens to fit the data better than the exact curve, so the evidence of the call rises. FPP then rises if the call is a false-positive scenario and falls if it is a planet scenario, as for TIC 424717049. This is the case of a planet call of TIC 403224147, of TIC 278659026 and of TIC 424717049. On the planet call of TIC 403224147 Δ is centred on zero (median -0.003) and the published code fits better on 47.5 percent of the 6,508 affected draws; the heaviest draw of the published code lies at the 99.92th percentile, where the published code fits better by 32.5, which is what a random error of the size of the noise and a maximum over draws would give. On the draw of Figure 2 (left), the spurious dip of the published code, about 0.07 d before the transit, falls on a dip of the data, structure that a likelihood with white noise does not model, and the favourable error is a fit to it. This is read from one draw of the planet call. On TIC 424717049 the distribution of Δ is centred on zero and the heaviest draw of the published code lies at the 99.99th percentile.

The second pattern is a large error of the published code on most draws of an eclipsing-binary call. In TIC 122609480, 405410684 and 340786513, the published code fits better on only 13 to 15 percent of the 4,084 to 12,573 affected draws of the call, with a tail of Δ to -1.6×10^6. The heaviest draw of the published code lies at the 91st to 98.5th percentile (Δ 11.2, 20.1 and 12.8). The error of the published code is large on most affected draws, and the draw that carries its evidence is one of the few on which it fits, so its evidence is the higher. In TIC 325177252, call 10, an eclipsing-binary call whose $\ln Z$ differs by $-40{,}351$, the heaviest draw is the same in both models ($k = 0.80$, $e = 0.26$, $a/R_\star = 2.46$, inclination 24.6°, period 0.927 d) and the error of the published code removes it, so the published code is lower; half the chi-squared against the data is

236,644 for the published code and 196,294 for the reference (Figure 2, right).

The third pattern is a spoiled exact draw. The error of the published code spoils the draw that carries the evidence of the exact model. This is TIC 42071226, where the heaviest draw of the published code agrees with the reference to 3×10^{-9} and on the heaviest draw of the exact model the published code fits worse by 131 in half chi-squared, so the published code is lower.

In the control call of a star that did not shift, the tail of Δ is as heavy (-1.9×10^{5}) and the heaviest draw is not an affected draw, so the published code and the exact model agree on it. A shift therefore needs the draw that carries a call to be an affected draw, and its sign depends on whether the error helps the fit of that call and on the scenario of the call. The first two patterns are read on the replayed stars only. How often each occurs for calls whose evidence rests on more draws, and the cause of the error inside PyTransit, are not shown here.

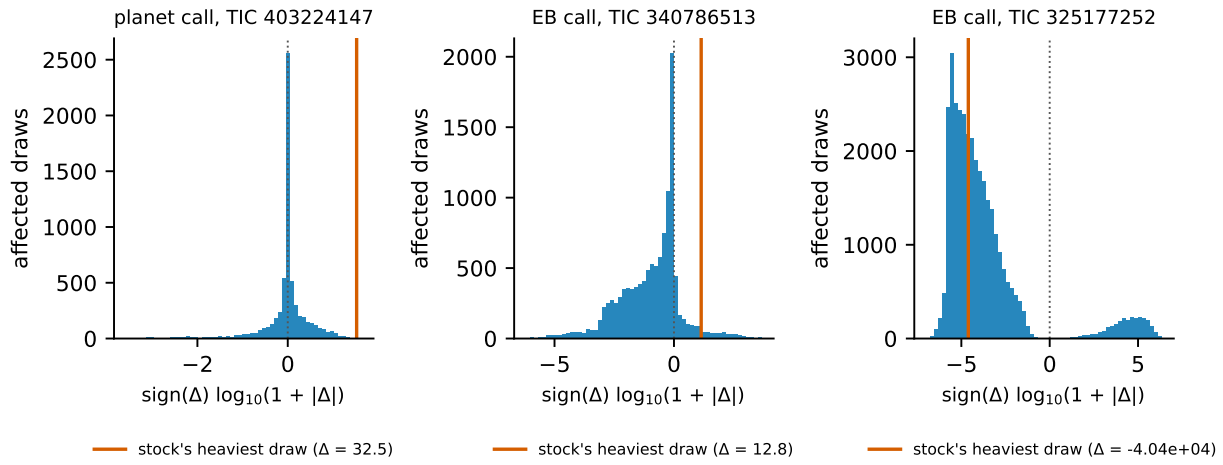


Figure 1: Distribution of Δ, half the chi-squared of the exact curve minus that of the published code's curve, over the affected draws of a planet call of TIC 403224147, an eclipsing-binary call of TIC 340786513 (the published code higher) and an eclipsing-binary call of TIC 325177252 (the published code lower), on a signed logarithmic scale, $\mathrm{sign}(\Delta)\log_{10}(1+|\Delta|)$. An affected draw is one on which the two curves differ by more than 10^{-6} at some point. Positive Δ means that the published code fits better. The vertical line is the published code's heaviest draw, the draw that carries the evidence of the call.

4.2 Attribution, and the shifts at 10^6 draws

TRICERATOPS 1.1.0 builds the `QuadraticModel` of PyTransit 2.2.0 with `interpolate=False` and calls its `evaluate_pv` for the light curve of each draw. The orbit of that model is a Taylor expansion about mid-transit. Parviainen and Korth [9] give its accuracy near transit: the maximum error is below 1 ppm for semi-major axes larger than 5 stellar radii and for "the major part of the physically plausible orbital parameter space", still below 10 ppm for very eccentric orbits, and below 1 ppm "for all but the shortest orbital periods and highest eccentricities"; their accuracy grid has periods of 2.5, 5, 15 and 30 d and $a/R_\star$ of 7.5 to 40. The draws that carry the shifts of Section 4.1 have periods of 0.60 to 1.23 d, all below the tested range, scaled semi-major axes of

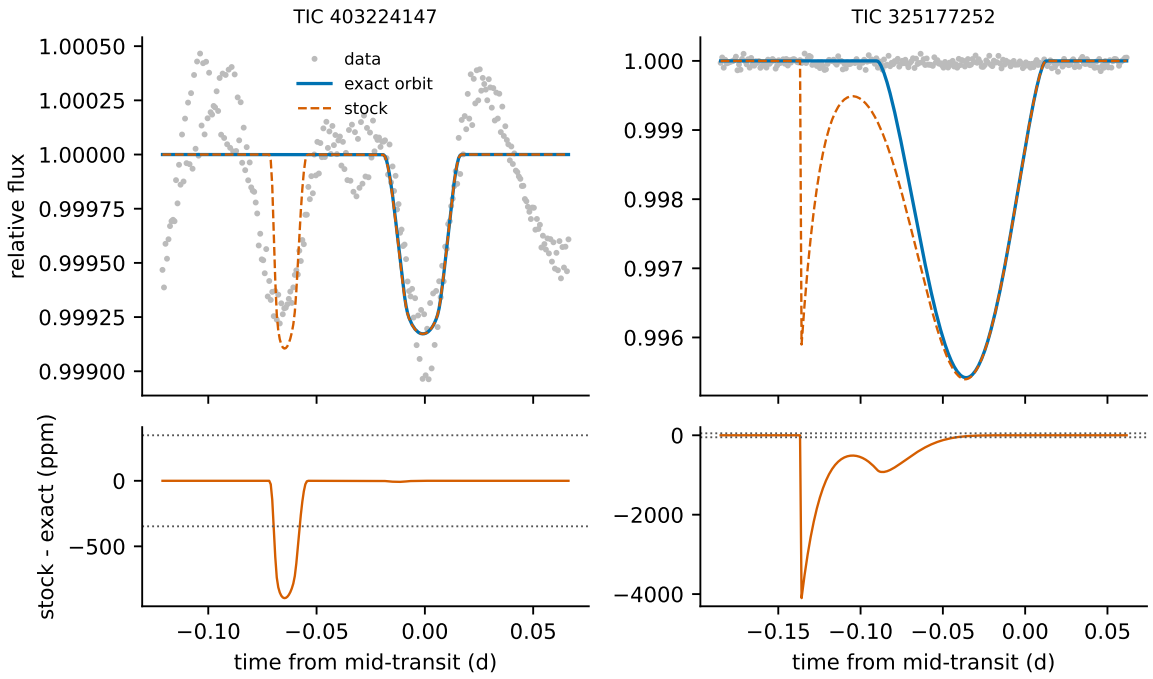


Figure 2: The light curve of the draw that the published code weights most in the planet call of TIC 403224147 (left) and in the eclipsing-binary call of TIC 325177252 (right): the data (points), the exact curve (solid) and the published code's curve (dashed), with their difference below. The data are the binned fold of 1,000 points that the calculation uses, not binned further; the dotted lines are the noise per point, about ±347 ppm on the left and ±50 ppm on the right. On the left the published code has a second dip of the full depth about 0.07 d before the transit, where the exact curve is flat.

2.3 to 12.1 (five of eight under 5), eccentricities of 0.26 to 0.80, and periastron distances $a(1-e)$ of 1.4 to 3.0 stellar radii. TRICERATOPS also evaluates the curve over the whole fold, up to half a period from mid-transit, far from the point of the expansion. We read the shifts as the use of the published model beyond the range in which its authors tested it, and state it so; we did not test PyTransit inside its range, and we do not call its model defective. The cause inside the model has not been shown. A replica of the expansion, written from the text of the paper and not from the code of PyTransit, reproduces the log-likelihood of the published code to four digits on three extreme draws of two stars (TIC 42071226 and 206361339), but not on the heavy draws of TIC 201354238, where the published curve differs from the exact one by 7.0×10^{-4} and 7.8×10^{-4} and the replica by 1.7×10^{-6} and 2.7×10^{-6}.

Collier et al. [10] and Giacalone et al. [1] run 10^6 draws. Table 3 gives FPP at that number on the same four stars, with three seeds and both calculations. The shift persists on TIC 278659026, where the FPP of the published code minus that of the default is +0.40 to +0.41, and on TIC 42071226, where it is −0.033 to −0.034, with a scatter of each calculation under 0.005 over the seeds. The side change persists on TIC 325177252: the published code stays under the line of 0.015 (0.0005 to 0.00075) and the default stays over it (0.030 to 0.032) at all three seeds. On TIC 424717049 the rule of persistence is not met: the published code gives 0.00066, 0.978 and 0.119 at its three seeds,

a mean of 0.37, and the default 0.872 to 0.876; the two are on opposite sides of 0.5 at seeds 1 and 3 and on the same side at seed 2. The FPP of the published code on that star depends on the seed at 10^6 draws, and that of the default does not.

Table 3: FPP at 10^6 draws, seeds 1 to 3, of the published code and of the default of the package on the same draws. The shift persists if the difference has one sign and is at least 0.01 at all three seeds; a side change persists if the two stay on opposite sides of the line at all three seeds.

TIC	published code	default	result
278659026	0.9999999785, 0.9999997808, 0.9999998249	0.5955, 0.5902, 0.5856	shift persists (+0.41)
42071226	0.7428, 0.7359, 0.7387	0.7760, 0.7702, 0.7729	shift persists (−0.034)
325177252	0.00051, 0.00075, 0.00052	0.0303, 0.0301, 0.0320	side change of 0.015 persists
424717049	0.00066, 0.978, 0.119	0.8752, 0.8756, 0.8722	rule not met

5 Performance

5.1 The card and the time of the calculation

The card's rates. On a Radeon RX 7900 XTX (ROCm 7.2) the arithmetic of one exposure node of the kernel (three small-angle Newton steps, the sky separation and the order-3 series) takes 0.534 ns per element in double precision and 0.036 ns in single precision, a ratio of 14.7. The published code on 32 CPU threads (AMD Ryzen 9 9950X3D) takes 0.89 ns per element at the same quadrature of 20 midpoints, so a double-precision kernel at the arithmetic floor of 0.53 ns would be at most about 1.7 times faster at that quadrature. No launch shape of 15 tried is faster than the default of 32 lanes in one warp, and changing the waves per execution unit from 1 to 8 changes the time by less than 1%.

The overlap and the elliptic integrals. With three scenario calls in flight the whole calculation is 1.45 to 1.71 times faster than with one, on four stars, with the same FPP at both settings. The Landen recurrence of the elliptic integrals took at most 7 of its 8 steps in 2.1 million calls.

What the single precision gains. Table 4 gives the time of the whole calculation on four stars, measured together with no other work on the device, medians of 5, the published code with its host-side lookups of the limb-darkening and stellar tables removed ("host fixes") as the reference. The device in double precision at 20 midpoints is 1.33 to 2.13 times faster than the reference, with the single-precision first pass 2.27 to 3.47 times, with 3 Gauss–Legendre nodes in double precision 2.76 to 6.76 times, and at the default (3 nodes with the single-precision first pass) 3.02 to 8.59 times. The host fixes make the published code 5% to 21% faster. The single-precision first pass cuts the time at 20 midpoints by 20% to 47% and not by the factor of 14.7 of the node loop, because the Kepler solve, the tier tests and the accumulation stay in double precision and, with the single-precision first pass, 14.7% (planet), 26.3% (eclipsing binary) and 33.0% (twin) of the pairs of the benchmark input need the closed form in double precision. With the node loop at the measured floor of 0.534 ns per element, the double-precision wall on the star with the most pairs flagged by

the double-precision first pass is at most 1.40 times the reference, and a separate measurement of that row on the same machine was within 6% of it.

Table 4: Seconds of the whole calculation (medians of 5) on one AMD Radeon RX 7900 XTX and a 32-thread AMD Ryzen 9 9950X3D, measured together, at 100,000 draws. The reference is the published code with the host fixes. The last four columns are the device at 20 midpoints and at 3 nodes, each in double precision and with the single-precision first pass; the last is the default of the package. The ratio of the reference time to the row is in parentheses.

TIC	published	reference	f64, 20 mid	f32, 20 mid	f64, 3 nodes	default
396212145	4.461	4.151	1.950 (2.13)	1.195 (3.47)	0.614 (6.76)	0.483 (8.59)
206361339	1.456	1.155	0.633 (1.82)	0.508 (2.27)	0.419 (2.76)	0.383 (3.02)
42071226	3.331	3.033	1.809 (1.68)	1.112 (2.73)	0.715 (4.24)	0.545 (5.57)
201354238	5.548	5.251	3.954 (1.33)	2.094 (2.51)	1.230 (4.27)	0.776 (6.77)

The three scenario likelihoods alone, on 100,000 draws and 1,000 points, take 5.11 s with the published code on 32 threads, 4.20 s on the device in double precision at 20 midpoints, 1.90 s with the single-precision first pass, 1.02 s at 3 nodes and 0.49 s at the default, 10.35 times less than the published code.

5.2 Twenty runs of 10^6 draws

Collier et al. [10] make 20 runs of 10^6 draws per target. Table 5 gives the time on one consumer card for the default of the package and for the published code on 32 CPU threads, on two stars. The published code was run at 10^5 and 2×10^5 draws, because its own guard refuses a batch of draws by points larger than 2 GB, and its time and memory at 10^6 draws are linear extrapolations of those two rows; its time doubles with the draws (1.97 and 2.00 times from 10^5 to 2×10^5 on the two stars). The default was run at 10^6 draws.

The default takes 88 s and 105 s for the 20 runs of 10^6 draws, 10.1 and 6.3 times less than the published code. The memory of the published code rises by 514 and 856 MiB from 10^5 to 2×10^5 draws, 0.5 to 0.8 GiB per 10^5 draws, so a run at 10^6 draws would fit in the memory of the machine; the default needs 2.7 GiB of host memory. One run of the published code at 10^6 draws was made, on a third star, TIC 325177252: 88.8 s and 11.5 GiB of resident memory, against the 33.1 and 44.6 s extrapolated for the two stars of the table, so the cost per star differs by about a factor of two on these three.

Device memory. Each likelihood call holds arrays of valid draws by points in double precision, and with three calls in flight these stack. The likelihoods run in chunks of at most 2^{26} draws times points, which bounds a call at 1.6 GiB. With the chunks, the device peak of the default at 10^6 draws is 4.1, 3.8, 4.2 and 3.9 GiB on TIC 396212145, 42071226, 325177252 and 424717049. Without the chunks the peak is 11.0 and 15.9 GiB on the first two, and on the other two the calculation runs out of memory at its first call on a card of 24 GiB, asking for 3.5 GiB with 21.3 to 21.5 GiB allocated. The chunks do not change the time: a calculation takes 4.432 and 5.267 s on the first two stars with them and 4.342 and 5.250 s without, measured together, and 12.3 and 9.1 s on the other two with them. The time depends on the star by a factor of about three at this number of draws. At 2×10^5 draws, where some calls are chunked, FPP, NFPP and all 42 scenario probabilities are

Table 5: Time and memory of 20 runs of 10^6 draws with the published code and with the default, on two stars, measured together after three quiet samples of the machine (medians of 3, 32 CPU threads for the published code, one AMD Radeon RX 7900 XTX with 24 GiB for the default). Measured: the published code at 10^5 and 2×10^5 draws and the default at 10^5, 2×10^5 and 10^6 draws. Extrapolated: the run of the published code at 10^6 draws, ten times its run at 10^5 (the two points of each star lie on a line through the origin to within 1 percent), the 20 runs, and its memory at 10^6 draws, on the line through its two points. The 20 runs of the default are 20 times its measured run. The ten batches are ten runs of 10^5 draws. Device memory, with the likelihoods run in chunks, is the peak allocated by torch (`max_memory_allocated`) over the whole invocation, which counts tensors only and not the reserved pool, the context or the compiler.

	TIC 396212145			TIC 42071226		
	published	default	ratio	published	default	ratio
one run, 10^5 draws (s)	4.455	0.492	9.1	3.307	0.547	6.0
one run, 2×10^5 draws (s)	8.793	0.941	9.3	6.603	1.052	6.3
one run, 10^6 draws (s)	44.6 (extrapolated)	4.408	10.1	33.1 (extrapolated)	5.237	6.3
20 runs, 10^6 draws (s)	891 (extrapolated)	88	10.1	661 (extrapolated)	105	6.3
ten batches of 10^5 draws (s)	44.6	4.92	9.1	33.1	5.47	6.0
resident memory, 10^5 draws (MiB)	1,975	1,881		2,275	1,891	
resident memory, 10^6 draws (GiB)	6.4 (extrapolated)	2.71		9.7 (extrapolated)	2.67	
device memory, 10^6 draws (GiB)		4.1			3.8	

equal to the last digit with and without the chunks, at three seeds, on TIC 325177252. In double precision, which the default does not use, the chunked likelihood differs from the whole in the last bit on about 1 percent of the draws, by up to 4×10^{-12} relative in the log-likelihood, because the double-precision flux of a draw depends in its last bit on the other draws of its launch; the default precision is bit for bit. A user who selects double precision should expect that difference.

6 The scatter between batches

The scatter of the probabilities between seeds depends on the star. At 10^5 draws the median scatter of FPP between ten seeds over 158 stars is 0.012, and 36 stars (0.23, Wilson 0.17 to 0.30) have a scatter over 0.05. On one star of the set, run at 10^6 draws with the published code, the FPP was 0.00066, 0.978 and 0.119 at three seeds. An error stated from the sample variance of the draws does not cover the truth where one draw dominates. On seven calls of four stars it covers it in 0.08 to 0.70 of runs, for a nominal 0.90.

velociraptor describes the scatter in another way. It runs a fixed number of seeded batches of N draws for each scenario call, pools them into the probabilities, and reports as the error of each probability the standard deviation between batches over the root of their number, and the effective sample size $(\sum w)^2 / \sum w^2$ of each call. The calls of a batch run one at a time (Section 2), and 200 of 200 evidences from 20 batches of each call, run twice at each of two settings, agree bit for bit. The coverage measurements below ran the calls of each batch one at a time. The coverage below was measured with batches of 10^5 draws, so ten batches hold 10^6 draws, the number of draws of one run of the published default. The coverage of a stated error is the share of runs whose interval contains the value that the error is stated for.

On the three stars where the batch count was chosen, the error covers at its nominal rate. The

measurement is 40 replicates of ten batches of 10^5 draws each per call, with the device calculation at 3 nodes in double precision, against the mean of 100 reference batches. The three stars have FPP of 0.02 and 0.04, near the validation line of 0.015, and 0.18. The error covers the truth in 0.925 to 0.975 of replicates for FPP and 0.90 to 0.95 for $\ln$ NFPP, against a nominal 0.90. The batch count was chosen on these stars, so the figures are in sample. The error does not cover where one call carries the variance. On the star whose leading call carries 0.99 of it, the coverage of $\ln$ NFPP is 0.525, because the truth rests on one rare batch that the ten batches miss together. Their spread is then small and the stated error too small, with a median of 0.79 against a root mean square error of 2.22, so the error gives no warning on that star.

Out of sample. Six further stars were taken by a rule that does not look at coverage: the first six in hash order whose FPP scatter over ten seeds exceeds 10^{-3}, excluding the four stars of the measurements above (the three and the one where a call carries the variance). Each ran 100 batches of 10^5 draws. Ten trials per star took a group of ten batches, the interval of ± 1.645 times the standard deviation over $\sqrt{10}$, and as the truth the pooled evidence of the other 90 batches. The interval contains the truth in 40 of 60 trials (0.667, Wilson 95 percent interval 0.54 to 0.77), by star 3, 10, 7, 8, 5 and 7 of 10. The interval lies entirely below the nominal 0.90, so the error taken from ten batches is not a valid 90 percent interval out of sample. Two checks made after the result, which decide nothing, do not repair it: Student's t quantile at 9 degrees of freedom gives 47 of 60 (0.783, 0.66 to 0.87), and groups of 20 batches against the other 80 give 23 of 30 (0.767, 0.59 to 0.88). The effective sample size of the leading calls in the first batch is 1.3 and 3.6 on the two stars that cover 5 and 3 of 10, and 7.6 to 63.7 on the four that cover 7 to 10 of 10. This is consistent with a stated error from few batches being small where the draws that carry the evidence are rare; six stars cannot establish it. The scatter between ten batches therefore describes the Monte Carlo scatter, and a class near a decision value is unsettled where the batches straddle it.

Use of the reported scatter. The package reports the scatter between ten batches as a description of the Monte Carlo scatter of a run and states no coverage for it.

7 Conclusions

The exact light curve changes the result of TRICERATOPS on a minority of stars. On 15 of 158 candidates the false-positive probability or the probability of a nearby star moves by at least one seed scatter, and on two of them FPP crosses a decision value: 0.015 on one, where the published code would validate the candidate and the exact calculation would not, and 0.5 on the other (Section 4.1). At 10^6 draws the shifts persist on three of the four stars tested, the change across 0.015 among them (Section 4.2).

The shifts come from the draws that carry the evidence of a scenario call. On those draws the light curve of PyTransit 2.2.0 is off a 50-digit reference by 4.5×10^{-4} to 4.1×10^{-3}, at periods of 0.60 to 1.23 d and eccentricities of 0.26 to 0.80, outside the range in which its authors tested it (Section 4.2). A candidate whose scenario draws reach short, eccentric orbits is the case in which the transit model decides the result.

On one consumer card, 20 runs of 10^6 draws take 88 s and 105 s on two stars, where the published code on 32 CPU threads would take an extrapolated 891 s and 661 s (Section 5). The single-precision first pass shifts the evidence by under 0.4 percent of its seed scatter on four stars (Section 3.3).

The scatter between ten seeded batches describes the Monte Carlo scatter of a run. It covered the truth at its nominal rate on the three stars on which the batch count was chosen and in 0.667 of trials on six stars chosen by a rule that does not look at coverage, so the package states no coverage for it (Section 6).

8 Limits and what was not verified

- The stars are hash-ordered samples of earlier calculations (Table 1). They are not draws from a catalog.
- The model shift is measured on the 158 stars of the screen set that were run (Table 1), candidates of a transit search and not stars in general or validated candidates. The published code ran at three seeds and the exact model at ten, so the shift is a mean over three paired seeds against the seed scatter of ten. The two changes of side are two cases and not a rate (Wilson interval of their share 0.35 to 4.5 percent). Two crowded fields, with more than 1,150 stars in their TRICERATOPS tables, are counted.
- The persistence at 10^6 draws is measured on four stars at three seeds. On TIC 424717049 the rule of persistence was not met, so the change of side of 0.5 on that star is not shown at that number.
- The attribution is an argument from the range in which the accuracy of the published model was tested. The accuracy of PyTransit inside that range was not tested here, and the cause inside the model has not been shown. The three patterns of Section 4.1 are read from the heaviest draw of the replayed calls of the shifted stars and of three controls. No pattern was tested on calls whose evidence rests on more draws, the secondary-eclipse tests inside the eclipsing-binary likelihoods were not compared, and TIC 278659026 was not replayed beyond its heaviest draws.
- PyTransit was compared at version 2.2.0 only, with `interpolate=False` as TRICERATOPS 1.1.0 calls it. No later release and no third light-curve code was run on the same draws.
- The coverage of the fixed batches is 0.925 to 0.975 in sample on three stars and 0.667 on six further stars; the dependence on the effective sample size rests on those six stars, fields with more than 177 stars in their tables were not measured, and no batch count that repairs the coverage was found. The scatter over many stars (Section 6) is measured on 158 stars of one earlier set of calculations at 10^5 draws, and describes that set and not TESS stars or validated candidates.
- No reliability curve on labelled candidates has been made, and agreement of the labels with other pipelines has not been checked. The candidates with a disposition among the 158 stars were not compared with their dispositions.

- The transit search that selected the 158 stars is not described beyond its role.
- The speed was measured on one consumer card (Radeon RX 7900 XTX) and one CPU. The ratio of the rates in single and double precision, and the conclusions about speed, hold for this card. Data-centre cards may have a very different ratio.
- The cost table is on two stars, medians of 3, measured together; the time and memory of the published code at 10^6 draws on those two stars are extrapolated from two rows, and no run of it at 10^6 draws was made on them. The comparison of the whole calculation with the published code rests on five repeats per row, measured together. Each calculation ran 100,000 draws per scenario, the number that the calling code fixes for these calculations.
- The practice of 20 runs of 10^6 draws per target is read from the text of one recent paper [10]; Giacalone et al. [1] made the 20 runs for their 12 validated objects only.
- The product and flux column of the light curve are recorded for 136 of the 159 stars of the screen set (SPOC 2-minute, PDCSAP flux, no version tag in the file name); the other 23, from two earlier runs made before those columns were recorded, were made with the default product of the same code, and their flux column is not recorded. The TESScut cutouts and the catalogue entries of the fields are as of the day each field was built and are not versioned, so the field of a star cannot be rebuilt identically. None of the 136 recorded stars used a full-frame light curve as its light curve, and the 23 others used the default product, a 2-minute one; the pixels of every field come from full-frame cutouts. For a period in which the service gave cutouts without two header keywords, the missing keywords were set to zero, which does not change the default aperture.
- The device memory is the peak allocated by torch; the reserved peak was not measured. The equality of the probabilities with and without the chunks was tested at 2×10^5 draws on one star; at 10^5 draws no call is chunked. The cause of the dependence of the double-precision flux on the other draws of its launch is not located.

Software

Python 3.12; NumPy 1.26 [11]; SciPy [12]; PyTorch 2.12 [13]; Triton 3.7 [14]; Numba 0.68.1 [15]; TRICERATOPS 1.1.0 [1, 17]; PyTransit 2.2.0 [8]; mpmath 1.4 [16] for the 50-digit references.

Data availability

The light curves of the targets are the 2-minute light curves of the Science Processing Operations Center (SPOC), PDCSAP flux with the default quality mask, of the Mikulski Archive for Space Telescopes (MAST), read from its public data bucket; the folded light curve that TRICERATOPS receives is made from them. The field of each calculation (the pixels around the target, the neighbouring stars and their catalogue entries) is built by TRICERATOPS itself, from cutouts of the TESS full-frame images of the same sectors obtained from the TESScut service, an aperture of 5 by 5 pixels centred on the target, the TESS Input Catalog and Gaia catalogues, and a Gaia background

population, as they were on the day of the first calculation of each star. The package, its tests and the validation scripts are at `https://github.com/mauri870/velociraptor`. The records of the measurements of this paper other than the benchmarks and the validation of the package are available from the author on request.

A The 158 stars of the model comparison

Table 6 lists the 158 stars of the comparison of Section 4.1. FPP is the mean over seeds 1 to 3 of the published code and of the exact model on the same draws. A star is shifted under the rule of Section 4.1, in FPP, in ln NFPP or in both.

Table 6: The stars of the model comparison. Stratum F is flagged by the screen of Table 1 and N is not. FPP is the mean over three seeds of the published code (pub.) and of the exact model (exact) on the same draws. A dagger marks the two stars whose FPP changes side of 0.015 or 0.5.

TIC	stratum	FPP pub.	FPP exact	shifted	TIC	stratum	FPP pub.	FPP exact	shifted
26584326	F	0.755	0.763		231694736	N	0.300	0.288	
32052779	F	1.000	1.000		233680140	N	> 0.999	> 0.999	
38516712	F	0.333	0.333		234291977	N	0.561	0.559	
40339576	F	0.214	0.214		237158650	N	0.694	0.694	
42071226	F	0.739	0.775	FPP	237325365	N	> 0.999	> 0.999	
122609268	F	1.000	1.000		243354593	N	> 0.999	> 0.999	
122609480	F	> 0.999	0.895	FPP	256865831	N	> 0.999	> 0.999	
150431617	F	> 0.999	> 0.999		259719464	N	0.246	0.246	
166732511	F	0.376	0.374		261526008	N	3.2×10^{-10}	1.3×10^{-10}	
167303776	F	1.000	1.000		262818577	N	0.556	0.557	
193318850	F	0.262	0.263		265438551	N	0.032	0.028	NFPP
197949562	F	0.049	0.049		265588613	N	0.045	0.045	
201233593	F	3.5×10^{-4}	3.5×10^{-4}		278659026	N	> 0.999	0.577	FPP, NFPP
201235927	F	0.251	0.251		292292315	N	1.000	1.000	
229796245	F	0.999	0.999		293419639	N	1.000	1.000	
231670778	F	1.000	1.000		294198241	N	0.438	0.452	
231910539	F	0.999	0.999		295615371	N	1.000	1.000	
238815210	F	0.333	0.333		296739893	N	0.997	0.997	
261460023	F	0.988	0.988		302107568	N	0.056	0.056	
288344202	F	1.000	1.000		311401333	N	0.141	0.141	
292542560	F	1.000	1.000		311893557	N	1.000	1.000	
294884819	F	1.000	1.000		314864866	N	0.999	0.999	
300010099	F	1.000	1.000		316787978	N	0.096	0.096	
302061421	F	1.000	1.000		316936115	N	0.881	0.880	
340786513	F	> 0.999	0.806	FPP	318566416	N	0.124	0.124	
343886954	F	0.208	0.207		321908598	N	1.000	1.000	
345088232	F	> 0.999	0.991		321908619	N	1.000	1.000	
355547387	F	1.000	1.000		325177252	N	8.6×10^{-4}	0.034	FPP†
360918475	F	1.000	1.000		325210265	N	0.946	0.940	
361281902	F	0.198	0.198		325338668	N	1.000	1.000	
382575967	F	1.000	1.000		327724364	N	0.359	0.359	
389021976	F	1.000	1.000		328100489	N	0.911	0.911	
389752750	F	1.000	1.000		340003212	N	1.000	1.000	
391903064	F	1.000	1.000		340878553	N	1.000	1.000	
394557413	F	1.000	1.000		341497604	N	0.945	0.945	
402546736	F	0	2.4×10^{-14}		343647193	N	1.000	1.000	
403224147	F	> 0.999	0.701	FPP	343887624	N	1.000	1.000	
405410684	F	> 0.999	0.885	FPP	347030355	N	0.179	0.178	
406322391	F	0.020	0.020		350222730	N	1.000	1.000	

TIC	stratum	FPP pub.	FPP exact	shifted	TIC	stratum	FPP pub.	FPP exact	shifted
410148949	F	1.000	1.000		354654467	N	1.000	1.000	
410214986	F	0	0		358516596	N	0.847	0.836	FPP
425884522	F	0.152	0.152		359497943	N	0.076	0.074	
425923664	F	1.000	1.000		360453460	N	1.000	1.000	
466094094	F	0.193	0.192		365660822	N	> 0.999	0.973	NFPP
2055541811	F	0.994	0.994		369964729	N	5.4×10^{-4}	5.4×10^{-4}	
13574905	N	0.871	0.873		373431424	N	> 0.999	> 0.999	
26826078	N	0.994	0.994		375506058	N	0.333	0.333	
55652896	N	0.667	0.667		376669357	N	1.000	1.000	
69679391	N	0	0		377292556	N	0.485	0.485	
69747919	N	0.465	0.443		382039146	N	0.574	0.599	NFPP
73644471	N	0.360	0.360		382384315	N	1.000	1.000	
101944811	N	0.107	0.107		391880567	N	0.706	0.707	
139370537	N	0.204	0.204		391949880	N	0.104	0.104	
140889223	N	0.989	0.989		392747437	N	0.790	0.790	
141975044	N	0.155	0.155		393977332	N	1.000	1.000	
146320244	N	0.096	0.096		396211656	N	1.000	1.000	
150391436	N	0.644	0.667	NFPP	396212145	N	1.000	1.000	
155896956	N	0.051	0.051		401827140	N	0.140	0.140	
158324245	N	0	0		402046099	N	1.000	1.000	
160220653	N	1.000	1.000		402972169	N	1.000	1.000	
161378663	N	0.104	0.104		403121375	N	0.015	0.014	
163029327	N	0.018	0.018		405411672	N	1.000	1.000	
165124221	N	0.214	0.199	FPP	407575998	N	0.006	0.006	
165145322	N	1.000	1.000		408310006	N	0.931	0.931	
169556484	N	0.524	0.524		410056850	N	0.005	0.005	
177253429	N	> 0.999	> 0.999		410118578	N	1.000	1.000	
179582003	N	0.996	0.996		410402714	N	0.279	0.279	
198354567	N	0.253	0.253		417679385	N	0.558	0.518	FPP
201252144	N	0.674	0.674		417931607	N	0.087	0.087	
201288535	N	1.000	1.000		421184129	N	0.017	0.017	
201289552	N	0.443	0.443		421472503	N	0.052	0.052	
201306147	N	0.646	0.646		424717049	N	0.104	0.877	FPP†
206288957	N	0.643	0.641		448714306	N	0.284	0.284	
206361339	N	0.268	0.268		448922221	N	1.000	1.000	
215480963	N	0.800	0.800		449114791	N	1.000	1.000	
219327284	N	0.190	0.190		466095512	N	1.000	1.000	
220021331	N	0.765	0.765		466131866	N	1.000	1.000	
229152539	N	0.082	0.082		466207143	N	1.000	1.000	
230086768	N	1.000	1.000		2026649904	N	> 0.999	> 0.999	

References

[1] Giacalone, S., Dressing, C. D., Jensen, E. L. N., et al. 2021, Vetting of 384 TESS Objects of Interest with TRICERATOPS and Statistical Validation of 12 Planet Candidates, AJ, 161, 24, arXiv:2002.00691

[2] Morton, T. D. 2012, An Efficient Automated Validation Procedure for Exoplanet Transit Candidates, ApJ, 761, 6, arXiv:1206.1568

[3] Morton, T. D., Bryson, S. T., Coughlin, J. L., et al. 2016, False Positive Probabilities for all Kepler Objects of Interest: 1284 Newly Validated Planets and 428 Likely False Positives, ApJ, 822, 86, arXiv:1605.02825

[4] Díaz, R. F., Almenara, J. M., Santerne, A., et al. 2014, PASTIS: Bayesian extrasolar planet validation. I. General framework, models, and performance, MNRAS, 441, 983, arXiv:1403.6725

[5] Torres, G., Fressin, F., Batalha, N. M., et al. 2011, Modeling Kepler transit light curves as false positives: Rejection of blend scenarios for Kepler-9, and validation of Kepler-9d, a super-Earth-size planet in a multiple system, ApJ, 727, 24, arXiv:1008.4393

[6] Mandel, K., & Agol, E. 2002, Analytic Lightcurves for Planetary Transit Searches, ApJ, 580, L171, arXiv:astro-ph/0210099

[7] Agol, E., Luger, R., & Foreman-Mackey, D. 2020, Analytic Planetary Transit Light Curves and Derivatives for Stars with Polynomial Limb Darkening, AJ, 159, 123

[8] Parviainen, H. 2015, MNRAS, 450, 3233

[9] Parviainen, H., & Korth, J. 2020, Going back to basics: accelerating exoplanet transit modelling using Taylor-series expansion of the orbital motion, MNRAS, 499, 3356, arXiv:2009.09965

[10] Collier, M., Giacalone, S., Derfer, B., et al. 2026, Assessing the Impact of High-Resolution Imaging on Statistical Validation of TESS Planet Candidates, arXiv:2608.21560

[11] Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Array programming with NumPy, Nature, 585, 357, doi:10.1038/s41586-020-2649-2

[12] Virtanen, P., et al. 2020, SciPy 1.0: fundamental algorithms for scientific computing in Python, Nature Methods, 17, 261, doi:10.1038/s41592-019-0686-2

[13] Paszke, A., et al. 2019, PyTorch: An Imperative Style, High-Performance Deep Learning Library, Advances in Neural Information Processing Systems 32, arXiv:1912.01703

[14] Tillet, P., Kung, H. T., & Cox, D. 2019, Triton: an intermediate language and compiler for tiled neural network computations, Proceedings of the 3rd ACM SIGPLAN International Workshop on Machine Learning and Programming Languages, 10, doi:10.1145/3315508.3329973

[15] Lam, S. K., Pitrou, A., & Seibert, S. 2015, Numba: a LLVM-based Python JIT compiler, Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC, doi:10.1145/2833157.2833162

[16] Johansson, F., et al., mpmath: a Python library for arbitrary-precision floating-point arithmetic, Zenodo, doi:10.5281/zenodo.1476882

[17] Giacalone, S., & Dressing, C. D. 2020, triceratops: Candidate exoplanet rating tool, Astrophysics Source Code Library, ascl:2002.004